Schedule

Agentic AI for Earth Observation

The workshop for Agentic AI and Earth Observation in Berlin, Germany, 19-21 October 2026

Day 1 • 19 October 2026

Time	Foyer & Wandelhalle Ground Floor & 1st Floor	Lecture Hall 2nd Floor	Expertisium Ground Floor	B. von Langenbeck 1st Floor	Werner-Körte 1st Floor	Rudolf-Virchow 2nd Floor
12:00 – 13:00	12:00–13:00 Registration & Welcome Lunch					
13:00 – 13:45		13:00–13:45 Opening Remarks				
13:45 – 14:30		13:45–14:30 Keynote: Agentic AI for Earth Observation and Climate Resilience Prof. Dr. Salman Khan Mohamed bin Zayed University of Artificial Intelligence				
14:30 – 15:00	14:30–15:00 Coffee Break | 4 Demos Foyer: Demo 13 & Demo 14 Wandelhalle: Demo 11 & Demo 15		14:30–15:00 Coffee Break | 2 Demos Demo 12 & Demo 17		14:30–15:00 Featured Demo 1	14:30–15:00 Featured Demo 2
15:00 – 16:00		15:00–16:00 Oral Session 1: Agentic Decision Intelligence and Earth Action 5 talks				
16:05 – 16:45		16:05–16:45 Panel: Earth Observation in the age of Agentic AI: From Insight to Business				
17:00 – 17:25		17:00–17:30 Thematic Discussion: Subway Takes: Agentic AI Edition		17:00–18:30 Poster Session 1 22 posters	17:00–17:25 Featured Demo 3	17:00–17:25 Featured Demo 4
17:25 – 17:30					17:25–17:50 Featured Demo 5	17:25–17:50 Featured Demo 6
17:30 – 17:50						
17:50 – 18:15					17:50–18:15 Featured Demo 7	17:50–18:15 Featured Demo 8
18:15 – 18:30					18:15–18:40 Featured Demo 9	18:15–18:40 Featured Demo 10
18:30 – 18:40	18:30–20:45 Welcome Reception					
18:40 – 20:45						

Day 2 • 20 October 2026

Time	**Foyer & Wandelhalle** Ground Floor & 1st Floor	**Lecture Hall** 2nd Floor	**Expertisium** Ground Floor	**B. von Langenbeck** 1st Floor	**Werner-Körte** 1st Floor	**Rudolf-Virchow** 2nd Floor
08:00 – 08:45	08:00–08:45 Welcome Coffee					
08:45 – 09:30		08:45–09:30 Keynote: The Rise of Agentic AI: Ecosystem, Trends and Implications for Science and Industry Dr. Jacopo Chevallard Mistral AI				
09:30 – 10:15		09:30–10:15 Keynote: FAME - Traditional and Agentic AI in Space Dr. Steve Chien Jet Propulsion Laboratory (JPL), NASA				
10:15 – 10:45	10:15–10:45 Coffee Break | 4 Demos Foyer: Demo 3 & Demo 5 Wandelhalle: Demo 20 & Demo 21		10:15–10:45 Coffee Break | 2 Demos Demo 8 & Demo 22		10:15–10:45 Featured Demo 11	10:15–10:45 Featured Demo 12
10:45 – 11:45		10:45–11:45 Oral Session 2: Autonomous EO Systems, Space Operations and Cognitive Computing in Space 5 talks				
12:00 – 12:30		12:00–12:30 Panel: From Earth Observation to Earth Intelligence: Opportunities, Challenges and Future Directions				
12:30 – 12:45		12:30–12:45 Group Photo Shoot				
12:45 – 14:00	12:45–14:00 Hosted Lunch					
14:00 – 14:30		14:00–14:30 Keynote: Failure Mode in Agentic Reasoning Prof. Dr. Manling Li Northwestern University & Amazon Scholar				
14:30 – 15:45		14:30–15:45 Oral Session 3: Design, Development and Evaluation of AI Agents 6 talks				
15:45 – 16:15	15:45–16:15 Coffee Break | 4 Demos Foyer: Demo 1 & Demo 7 Wandelhalle: Demo 2 & Demo 4		15:45–16:15 Coffee Break | 2 Demos Demo 6 & Demo 9		15:45–16:15 Featured Demo 13	15:45–16:15 Featured Demo 14
16:15 – 17:15		16:15–17:15 Technical deep dives: Building with Mistral Open Agentic AI tools: From Scientific Discovery to Operational Services				
17:15 – 17:45		17:15–17:45 Keynote: Harnessing the Collective Intelligence of AI Agents for Discoveries Prof. Dr. James Zou Stanford University				
17:45 – 18:10				17:45–19:00 Poster Session 2 21 posters	17:45–18:10 Featured Demo 15	17:45–18:10 Featured Demo 16
18:10 – 18:35					18:10–18:35 Featured Demo 17	18:10–18:35 Featured Demo 18
18:35 – 19:00					18:35–19:00 Featured Demo 19	18:35–19:00 Featured Demo 20

Day 3 • 21 October 2026

Time	**Foyer & Wandelhalle** Ground Floor & 1st Floor	**Lecture Hall** 2nd Floor	**Expertisium** Ground Floor	**B. von Langenbeck** 1st Floor	**Werner-Körte** 1st Floor	**Rudolf-Virchow** 2nd Floor
08:00 – 09:00	08:00–09:00 Welcome Coffee					
09:00 – 09:45		09:00–09:45 Keynote: Building Robust Economic Capabilities in AI Agents Davide Crapis Ethereum Foundation				
09:45 – 10:30		09:45–10:30 Keynote: Why Mature Agents Require a Paradigm Shift in Tooling Dr. Christopher Forrest Brown Google DeepMind				
10:30 – 11:00	10:30–11:00 Coffee Break | 2 Demos Foyer: Demo 18 & Demo 19		10:30–11:00 Coffee Break | 2 Demos Demo 10 & Demo 16		10:30–11:00 Featured Demo 21	10:30–11:00 Featured Demo 22
11:00 – 12:00		11:00–12:00 Oral Session 4: Multi-Modal Large Models, Datasets and Agentic Earth Intelligence 5 talks				
12:15 – 12:30		12:15–12:30 Closing Ceremony				
12:30 – 14:00	12:30–14:00 Hosted Lunch					
14:00 – 15:30			14:00–15:30 Track 1: Building and Operating EO Agents - Building Agentic Earth Observation Workflows with EVE Workbench Andreas Haida, Bara Safrankova - Frank Reply	14:00–15:30 Track 2: Trustworthy and Scientific Agentic AI - How Do You Know Your Agent Works? Evaluating Earth Observation Agents with Existing Benchmarks Àlex R. Atrio, Antonio Lopez, Jino Rohit - Pi School, Nicolas Longépé - ESA labs		14:00–15:30 Track 3: Foundation Models, Embeddings and Lightweight AI (Part 1) - Small Models, Big Maps - Local Geospatial Workflows with Fine-tuned SLMs Soumya Ranjan, Daniel Wiesmann - Development Seed
15:30 – 16:00	15:30–16:00 Coffee Break					
16:00 – 17:30			16:00–17:30 Track 1: Building and Operating EO Agents - Agentic Analysis of Earth Observation for Utility Intelligence Fuxun Yu, Rishi Madhok - TerraByte AI	16:00–17:30 Track 2: Trustworthy and Scientific Agentic AI - From Vibe Coding to Verifiable Scientific Discovery: Agentic Workflows for Earth Observation Rocco Sedona, Pinaki Banerjee, Alexandre Strube - Forschungszentrum Jülich, Rohit Sahoo - The University of Alabama in Huntsville, NASA Marshall Space Flight Center		16:00–17:30 Track 3: Foundation Models, Embeddings and Lightweight AI (Part 2) - Small Models, Big Maps - Local Geospatial Workflows with Fine-tuned SLMs Soumya Ranjan, Daniel Wiesmann - Development Seed

Detailed Programme

Day 1 • 19 October 2026

12:00–13:00 *Registration & Welcome Lunch*
Foyer & Wandelhalle

13:00–13:45 Opening Remarks
Lecture Hall

Speakers:

- **Wolfgang Richter** Moderator
- **Dorothee Bär** Federal Minister of Research, Technology and Space (Video)
- **Dr. Henry Marx** Berlin State Secretary for Higher Education and Research
- **Prof. Dr. Fatma Deniz** President of TU Berlin
- **Prof. Dr. Volker Markl** BIFOLD Director
- **Dr. Claudio Iacopino** Head of ESA Φ-lab Explore Office
- **Prof. Dr. Begüm Demir** BIFOLD Deputy Director & **Dr. Nicolas Longépé** EO Data Scientist at ESA Φ-lab Explore Office

13:45–14:30 Keynote: Agentic AI for Earth Observation and Climate Resilience
Lecture Hall • *Prof. Dr. Salman Khan*
Mohamed bin Zayed University of Artificial Intelligence

14:30–15:00 *Coffee Break | 4 Demos*
Foyer: Demo 13 & Demo 14
Wandelhalle: Demo 11 & Demo 15
Foyer & Wandelhalle

Demo 13 • Foyer

From Chat to Copernicus: Agent-Based Composition of Reproducible openEO Workflows on CDSE
Hans Vanrompay, Victor Verhaert, Mathias Michielsen • VITO

Demo 14 • Foyer

SAI4EO: From Conversational Remote-Sensing Tool Use to Agentic, Artifact-Grounded Geospatial Intelligence
Nicolò Taggi • Planetek Italia

Demo 11 • Wandelhalle

EVE Workbench Live: A Multi-Agent Debate on Berlin's Climate Stress
Daniel Fitzpatrick, Maria Müller • Frank Reply

Demo 15 • Wandelhalle

Portable Agentic openEO: From Natural Language to Executable Earth Observation Workflows
Suriyah Dhinakaran • EURAC Research Italy

14:30–15:00 *Coffee Break | 2 Demos*
Demo 12 & Demo 17
Expertisium

Demo 12

DEA Next: Making Earth Intelligence Conversational through Agentic AI
Arturo Montieri • Alia Space Systems

Demo 17

terraAI: Conversational Earth Observation Analysis

Evgenij Shapovalov, Artiom Hovhannisyan, Valentas Gružauskas • Vilnius University

14:30–15:00 Featured Demo 1

Werner-Körte

Earthward: A Human-in-the-Loop Architecture for Traceable, Explainable Agentic EO Workflows - Earthward

Presented by Sudhakaran Chithamanan, Sahil, Srinivas Sakthiaruna

14:30–15:00 Featured Demo 2

Rudolf-Virchow

Saferplaces Agentic AI - An Agentic Geospatial Copilot for Flood Risk Intelligence and Map-Based Decision Support - GECOsistema SRL

Presented by Tommaso Redaelli

15:00–16:00 Oral Session 1: Agentic Decision Intelligence and Earth Action

Lecture Hall

Chair: Dr. Anca Anghelea, ESA

This session focuses on agents that transform EO observations into actionable decisions. This session illustrates the evolution: from Observation, analysis, prioritization, decision support and Earth Action.

15:00–15:12

A Decision-Driven Hazard-Agnostic Agentic AI Framework for Earth Observation: Demonstration through a Flood Response Use Case

Firoozeh Foroozan, Carlos Pino Padilla • *Indra Space*

15:12–15:24

Multi-Agent Early Extreme Event Detection From IoT to Earth Observation Tasking

Arnaud Le Carvennec, Axel Gruyez • *CS Group*

15:24–15:36

Wildfire Lifecycle Monitoring from Earth Observation

Rishi Madhok, Fuxun Yu • *TerraByte AI*

15:36–15:48

Agentic Orchestration of Earth Observation Foundation Models via the Model Context Protocol

Pinaki Banerjee, Rocco Sedona • *Jülich Supercomputing Centre, Forschungszentrum Jülich, Germany*

15:48–16:00

TerraScope: Pixel-Grounded Visual Reasoning for Earth Observation

Yan Shu • *University of Trento*

16:05–16:45 Panel: Earth Observation in the age of Agentic AI: From Insight to Business

Lecture Hall

From data to insight, and from insight to action: The panel will explore how Agentic AI could fundamentally reshape the EO value chain, from sensing and data generation to analysis, decision-making and business impact. Bringing together perspectives from research, EO technology and enterprise, the discussion will examine how AI agents could be a key element of the full data chain, combine EO with diverse sources of knowledge, and transform increasingly rich Earth observations into decision-ready intelligence. With applications ranging from insurance and risk management to emerging EO services and markets, the panel will address the key question: How do we move from technological potential to real-world impact and sustainable business value?

Moderated by Chiara Cocchiara, ESA Φ-lab

Panelists:

- **Julia Gottfriedsen** Chief Technical Officer at ConstellR

- **Marcin Detyniecki** Group Chief Data Scientist & Head of Research & Advanced AI at AXA
- **Andres Camero** Co-Head of Department EO Data Science at German Aerospace Center (DLR)
- **Linda Sheard** VP, Analytics and Agentic Products at Xoople

17:00–17:30 Thematic Discussion: Subway Takes: Agentic AI Edition

Lecture Hall

Join this guided, engaging discussion where participants share and debate strong opinions, concerns, and unpopular takes on Agentic AI for Earth Observation. Come with a take of your own, or respond to one raised by someone else. The goal is not consensus, but a better collective understanding of where agentic approaches genuinely add value in EO.

Moderated by Eva Gmelich Meijling and Riccardo D'Ercole, ESA Φ-lab

17:00–18:30 Poster Session 1

B. von Langenbeck

Presenters stand by their boards for the full session.

Fitness for Purpose as an Agentic Task: Domain Encoded Reasoning for EO Data Selection in Regulated Environmental Work

Nahid Alemi Kermani • *Aniterra Pty Ltd*

SENSOR2RAG: Multimodal Sensor-to-RAG Architecture for Site-Specific Decision Support in Natural Capital Risk Management

Jacopo Soldateschi • *The Landbanking Group GmbH*

Beyond Task Success: Domain-Grounded Evaluation of Earth Observation Agents in Environmental Risk Workflows

Alexandros Oikonomidis • *Centre for Research and Technology Hellas (CERTH) and National and Technical University of Athens (NTUA)*

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Jaya Kumari • *Vortx AI*

CLUES: A Comprehensive Workflow for Integrating Geospatial Data in Biomedical Research

Marcel Jentsch • *Center of Digital Health BIH@Charité*

GEODES Assistant

Killian Perriot • *THALES*

From Isolated Models to Agentic Orchestration: A Ground Segment Multi-Agent Framework for Coastal and In-Land Water Resilience

Roger Escolà, Mariona Badenas-Agusti • *isardSAT*

Thematic Datacubes On-Demand (THEDE): An Agentic AI Framework for On-Demand EO Data Dissemination Using Large Language Models

Edoardo Kimani Bellotto • *Sistema GmbH*

EIS: An Open-Source Geo-Agent Framework Democratizing Environmental Knowledge and Collaboration

Matteo Mattiuzzi • *European Environment Agency*

AI Agentic Development of xAI for Earth Observation: Building SegExplainer
Yulia Grushetskaya, Mike Sips • *GFZ Helmholtz Centre for Geosciences*

Ask, Detect, Explain: An End-to-End Agentic System for Satellite Change Detection
Bartłomiej Ostrowski, Marcin Kluczek, Bartosz Augustyn, Jędrzej S. Bojanowski • *CloudFerro S.A.*

Agentic Interaction between Raster Flood Observations and GIS Features for Crisis Situational Awareness
Jente Bosmans • *VITO*

LATINSIGHT: A React-Based Multi-Agent System for Urban EO Analytics
Edoardo Schettino, Paolo De Piano • *Latitudo 40 S.r.l.*

ECHOSAT: Physically Consistent Spatio-Temporal Canopy Height Mapping as an Enabling Layer for Agentic Earth Observation
Berkant Turan • *Zuse Institute Berlin*

Engineering a Trustworthy NASA Earth Science Discovery Agent Using the CARE Methodology
Lillianne Thomas • *Development Seed*

A Simulation Study of LLM-Based Onboard Orchestration for Coordinated Earth-Observation Constellations
Christopher Castro Traba

Agentic Ground Segment: How Generative AI Redesigns Data Discovery and Processing, Ordering, and Operations
Filippo Bianchini • *Telespazio S.p.A.*

Hierarchical System with Agentic Refinement for Large-Scale Earth Observation Archives
Omjyoti Dutta, Joao Vitorino • *GMV S.L.*

AI-Driven ASAP Country Assessments
Filip Sabo • *European Commission, Joint Research Centre (JRC)*

From Earth Observation Data to Decision: An Agentic Architecture for Disaster Resilience
Bartosz Augustyn, Marcin Kluczek • *CloudFerro S.A.*

EO Agents in Actions - Lessons Learned from Building Products Around Analysis Agents
Lillianne Thomas, Drishtie Patel • *Development Seed*

GeoGuard: An Agentic Guardrails and Validation Framework for Geospatial AI
Rohit Sisir Sahoo • *The University of Alabama in Huntsville*

17:00 – 17:25 Featured Demo 3
Werner-Körte
Virtual Support for Earth Observation (VISEO): Agentic AI for Earth Observation Discovery – Thales Services Toulouse
Presented by Dorian Voydie, Killian Perriot, Romain Sixt

17:00 – 17:25 Featured Demo 4
Rudolf-Virchow
An Agentic Digital Twin Framework for Flood Risk Assessment:

Integrating Complex Workflows with Generative AI: From Satellite Acquisition to Risk Management – CS Group
Presented by Pauline Delporte, Yoann Steux

17:25 – 17:50 Featured Demo 5
Werner-Körte
WRI's Global Nature Watch: Geospatial AI in Plain Language – Development Seed Lisbon
Presented by Daniel Wiesmann, Soumya Ranjan, Leo Thomas

17:25 – 17:50 Featured Demo 6
Rudolf-Virchow
Max-ICS V2.0: Toward an Agentic EO Orchestration Platform – EarthLab Luxembourg S.A.
Presented by Mohamed Boukhebouze

17:50 – 18:15 Featured Demo 7
Werner-Körte
From Data & Pixels to Risk Context: Agentic Earth Observation for Explainable Climate Intelligence
Presented by Claudio Giorgio Giancaterino

17:50 – 18:15 Featured Demo 8
Rudolf-Virchow
GEONOVA: An Agentic EO Copilot for Geospatial Embeddings and Downstream Model Adaptation - Geonova P.C.
Presented by George Arvanitakis, Georgios Tsimiklis, Elisavet Parselia, Eleftheria Tetoula Tsonga, Dionysos Kampouris, Christos Tsekes

18:15 – 18:40 Featured Demo 9
Werner-Körte
Axis Spatial Migration Engine: From Fragmented EO/GIS Projects to Verified Pipelines - Axis Spatial
Presented by Najah Pokkiri

18:15 – 18:40 Featured Demo 10
Rudolf-Virchow
BlazerOps: Reinforcement Learning Environment for Accelerated Spatial ML Development - hyades.space
Presented by Ashin Alex, Harshil Magan

18:30 – 20:45 *Welcome Reception*
Foyer & Wandelhalle

Day 2 • 20 October 2026

08:00–08:45 *Welcome Coffee*
Foyer & Wandelhalle

08:45–09:30 Keynote: The Rise of Agentic AI: Ecosystem, Trends and Implications for Science and Industry
Lecture Hall • *Dr. Jacopo Chevallard*
Mistral AI
Chair: Dr. Nicolas Longépé, ESA Φ-lab

09:30–10:15 Keynote: FAME - Traditional and Agentic AI in Space
Lecture Hall • *Dr. Steve Chien*
Jet Propulsion Laboratory (JPL), NASA
Chair: Dr. Nicolas Longépé, ESA Φ-lab

10:15–10:45 *Coffee Break | 4 Demos*
Foyer: Demo 3 & Demo 5
Wandelhalle: Demo 20 & Demo 21
Foyer & Wandelhalle

Demo 3 • Foyer

Virtual Support for Earth Observation (VISEO): Agentic AI for Earth Observation Discovery
Dorian Voydie, Killian Perriot, Romain Sixt • Thales Services Toulouse

Demo 5 • Foyer

WRI's Global Nature Watch: Geospatial AI in Plain Language
Daniel Wiesmann, Soumya Ranjan, Leo Thomas • Development Seed Lisbon

Demo 20 • Wandelhalle

FLAP-0: An Interactive Platform for Agentic Earth Observation
Leonidas Kosmidis, Iker Fernandez Andonegui, Marc Solé i Bonet • Barcelona Supercomputing Center and Universitat Politècnica de Catalunya

Demo 21 • Wandelhalle

A Live Tour of EVE: An Agentic EO Platform Built on an Open MCP Tool Registry
Àlex R. Atrio, Antonio Lopez, Jino Rohit, Emanuele Micieli • Pi School

10:15–10:45 *Coffee Break | 2 Demos*
Demo 8 & Demo 22
Expertisium

Demo 8

GEONOVA: An Agentic EO Copilot for Geospatial Embeddings and Downstream Model Adaptation
George Arvanitakis, Georgios Tsimiklis, Elisavet Parselia, Eleftheria Tetoula Tsonga, Dionysos Kampouris, Christos Tsekes • Geonova P.C.

Demo 22

Agentic AI-Enabled Climate Downscaling with NASA's Prithvi WxC Foundation Model
Hugo Lee, Aashish Panta • JPL/NASA, University of Utah

10:15–10:45 Featured Demo 11
Werner-Körte
EVE Workbench Live: A Multi-Agent Debate on Berlin's Climate Stress

– Frank Reply
Presented by Daniel Fitzpatrick, Maria Müller

10:15–10:45 Featured Demo 12

Rudolf-Virchow

DEA Next: Making Earth Intelligence Conversational through Agentic AI – Alia Space Systems
Presented by Arturo Montieri

10:45–11:45 Oral Session 2: Autonomous EO Systems, Space Operations and Cognitive Computing in Space

Lecture Hall

Chair: Dr. Steve Chien, JPL/NASA
This session represents the most visionary part of the programme, around Cognitive Cloud Computing in Space (3CS), EO Virtual Constellations and autonomous missions.
"How do we move agentic EO from the cloud to autonomous constellations operating in orbit?"

10:45–10:57

Agentic Orchestration of Satellite Tasking and Data Retrieval in Earth Observation Virtual Constellations
Marie Leconte, Arnaud Le Carvennec, Pauline Delporte • *CS Group*

10:57–11:09

Autonomous Agentic Command and Control for Satellite Constellations: An AI-Accelerated Engineering Demonstration
Alessio Auddino • *Thales Alenia Space Italia*

11:09–11:21

Towards Cognitive Cloud Computing in Space: AI-eXpress as an Orbital Testbed for Agentic Earth Observation
Marco Mucci Beltrami • *Planetek Italia*

11:21–11:33

Agentic Mission Computing for Autonomous Earth Observation
Dave Ojika, Srinidhi Magesh (Flapmax), Leonidas Kosmidis, Iker Fernandez Andonegui, Marc Solé i Bonet, Eric Rufart Blasco (Barcelona Supercomputing Center and Universitat Politècnica de Catalunya), Projjal Gupta, Herman Lam (University of Florida), Aidan Osuch (University of California, Los Angeles), Eric Lim, Isabella Pier, Ben Cragin (University of Southern California)

11:33–11:45

Closing the Reasoning Latency Gap: FPGA-Accelerated LLM Agents for Onboard Supervision of Earth Observation Acquisition
Winfred Ooh-Azlin (Technische Universität Berlin), Pao Ying Heng (Royal Astronomical Society)

12:00–12:30 Panel: From Earth Observation to Earth Intelligence: Opportunities, Challenges and Future Directions

Lecture Hall

Moderated by Prof. Dr. Begüm Demir

Panelists:

- **Prof. Dr. Ioannis Papoutsis** National Technical University of Athens
- **Dr. Bertrand Le Saux** DG Connect at the European Commission
- **Mikhail Klassen** Planet Labs

12:30–12:45 *Group Photo Shoot*

Lecture Hall

12:45–14:00 *Hosted Lunch*

Foyer & Wandelhalle

14:00–14:30 Keynote: Failure Mode in Agentic Reasoning

Lecture Hall • *Prof. Dr. Manling Li*
Northwestern University & Amazon Scholar

Chair: Dr. Bertrand Le Saux, European Commission

14:30–15:45 Oral Session 3: Design, Development and Evaluation of AI Agents

Lecture Hall

Chair: Dr. Bertrand Le Saux, European Commission
This session addresses perhaps the most critical scientific question emerging from the workshop: "How should EO agents be built, evaluated, validated and trusted?".
This session is likely to generate discussion about evaluation, reproducibility, hallucinations, validation, governance, or trust.

14:30–14:42

Collaborative Agent Reasoning Engineering (CARE): A Structured, Context-Engineering Methodology for Systematically Designing Traceable and Maintainable AI Agents for Science
Rohit Sisir Sahoo • *The University of Alabama in Huntsville*

14:42–14:54

Evaluating EVE, an EO-Specialized LLM, in Agentic Earth Observation Systems
Riccardo D'Ercole, Eva Gmelich Meijling • *ESA Φ-lab*

14:54–15:06

EO Agent Arena: From Fragmented Benchmarks to Reproducible Evaluation of Agentic Earth Observation Systems
Najah Pokkiri • *Axis Spatial*

15:06–15:18

Behavioural Self-Monitoring for Autonomous Earth Observation Agents: The Attribute Reliability Monitor (ARM)
Alejandro Mousist • *Thales Alenia Space*

15:18–15:30

Balancing Accessibility and Scientific Validity in the Era of AI Agents and Climate Change
Elena Di Medio, Laia Romero, Guillermo Grau • *Lobelia Earth*

15:33–15:45

From Tools to Ecosystem: EVE's Community-driven MCP Registry for an Agentic Platform for Earth Intelligence
Àlex R. Atrio, Antonio Lopez, Jino Rohit, Emanuele Micieli • *Pi School*

15:45–16:15 *Coffee Break | 4 Demos*
Foyer: Demo 1 & Demo 7
Wandelhalle: Demo 2 & Demo 4
Foyer & Wandelhalle

Demo 1 • Foyer

Earthward: A Human-in-the-Loop Architecture for Traceable, Explainable Agentic EO Workflows
Sudhakaran Chithamanan, Sahil, Srinivas Sakthiaruna • Earthward

Demo 7 • Foyer

From Data & Pixels to Risk Context: Agentic Earth Observation for Explainable Climate Intelligence
Claudio Giorgio Giancaterino • Independent

Demo 2 • Wandelhalle

Saferplaces Agentic AI - An Agentic Geospatial Copilot for Flood Risk Intelligence and Map-Based Decision Support

Tommaso Redaelli • GECOsistema SRL

Demo 4 • Wandelhalle

An Agentic Digital Twin Framework for Flood Risk Assessment: Integrating Complex Workflows with Generative AI: From Satellite Acquisition to Risk Management

Pauline Delporte, Yoann Steux • CS Group

15:45 – 16:15 *Coffee Break | 2 Demos*

Demo 6 & Demo 9

Expertisium

Demo 6

Max-ICS V2.0: Toward an Agentic EO Orchestration Platform

Mohamed Boukhebouze • EarthLab Luxembourg S.A.

Demo 9

Axis Spatial Migration Engine: From Fragmented EO/ GIS Projects to Verified Pipelines

Najah Pokkiri • Axis Spatial

15:45 – 16:15 Featured Demo 13

Werner-Körte

From Chat to Copernicus: Agent-Based Composition of Reproducible openEO Workflows on CDSE – VITO

Presented by Hans Vanrompay, Victor Verhaert, Mathias Michielsen

15:45 – 16:15 Featured Demo 14

Rudolf-Virchow

SAI4EO: From Conversational Remote-Sensing Tool Use to Agentic, Artifact-Grounded Geospatial Intelligence – Planetek Italia

Presented by Nicolò Taggio

16:15 – 17:15 Technical deep dives: Building with Mistral Open Agentic AI tools: From Scientific Discovery to Operational Services

Lecture Hall

Chair: Prof. Dr. Ioannis Papoutsis, National Technical University of Athens

17:15 – 17:45 Keynote: Harnessing the Collective Intelligence of AI Agents for Discoveries

Lecture Hall • *Prof. Dr. James Zou*

Stanford University

Chair: Prof. Dr. Ioannis Papoutsis, National Technical University of Athens

17:45 – 19:00 Poster Session 2

B. von Langenbeck

Presenters stand by their boards for the full session.

Autonomous Geospatial Intelligence: An Agentic Workflow for End-to-End Feature Interpretation And Large-Scale Scene Monitoring

Marie-Charlotte Teulières • *Airbus Defense and Space Geo*

EOMAS - A Multi Agent System-Backed Assistant for Earth Observation

Hans Meine • *Fraunhofer Institute for Digital Medicine MEVIS and OHB Digital Services*

An Interactive Multi-Agent System for User-Driven And Context-Aware Hyperspectral Image Compression

Tristan Reimann, Maxim Smirnov, Viktor Tevosyan • *Technische Universität Berlin and BIFOLD*

Towards Earth Observation-Based Siting of Sponge-City Infrastructure in East African Cities: An Open-Data Framework

Karl A. W. Amolo • *Urban Canopy Kenya and East African Sponge Cities Initiative (EASCI)*

AI-Driven End-To-End (E2E) Monitoring in the Context Of The ESA Earth Observation Operations Framework Copernicus Space Component (EOF-CSC)

Caroline Ball • *Capgemini (France)*

From Perception to Priorities: An Agentic Pipeline for Multi-Hazard Triage and Explainable Resource Allocation in Earth Observation

Alessandro Princi • *RINA Consulting S.p.A*

Learned Satellite Embeddings as Covariates for Global Digital Soil Mapping: A Multi-Model, Multi-Resolution Evaluation Developed with AI Coding Agents

Laura Poggio • *ISRIC – World Soil Information*

Specialized Geospatial Foundation Models for Agentic Earth Observation: Lessons from Spheer FM

Mark Boer, Jordy de Lange • *Spheer*

Multi-Satellite Task Orchestration for Wildfire Monitoring with On-Board Visual-Language Models

Riccardo D'Ercole (ESA Φ-lab), Ana-Maria Pelin (Φ-lab and TU Delft)

Provenance-First Geospatial Composition: A Scaffold-Based Builder as a Trust-Bearing Counterpoint to Agentic Autonomy in Earth Observation

Sudhakaran Chithamanan (Earthward, 33 Blueprints Collective), Aravindh S. Subramanian (Technical University of Munich, Earthward)

Beyond the State of the Art in Geospatial Foundation Models: A Diagnostic Comparison of TerraMind and THOR

Frederick Schindlegger • *Bertelsmann Foundation*

Agentic AI as an Auditable Co-Scientist for Earth-Observation-Based Irrigation Mapping and Groundwater Accounting

Peter ReVelle • *Desert Research Institute (DRI)*

SeismicAgent: Benchmarking Evidence-Conflict Reasoning for Post-Earthquake Building-Damage Assessment

Fatma Elik • *Istanbul Metropolitan Municipality, Earthquake and Ground Investigation Directorate*

Towards Agentic AI Workflows for Earth Observation-Based Air Quality Studies

Alberto Vavassori, Yasil Yordanov • *Politecnico di Milano*

Enabling Adaptive Agentic Earth Observation across the Space–Ground–Cloud Continuum

Cecilia Calavaro • *Department of Civil Engineering and Computer Science Engineering, University of Rome Tor Vergata*

Manteo AI: Agentic Geospatial Intelligence for Earth Observation
Ioannis Prapas • *manteo.ai*

Airbus Geo Explore: Leveraging Semantic Search APIs for LLM Agentic Workflows
Fahd Benatia • *Airbus DS*

Aperture: Semantic Change Understanding and Grounded GeoAI Reasoning via SAR-Optical-Language Alignment
Magdalena Kaiser, Paul Weinmann • *LiveEO*

FlyPix AI: Agentic Vision-Language System
Ivan Tankoyeu • *Flypix AI*

Virtual Support for Earth Observation (VISEO)
Romain Sixt, Killian Perriot, Dorian Voydie • *THALES*

Advancing Disaster Resilience through Agentic AI Integration in the EO-PIN Suite
Guy Schumann • *RSS-Hydro*

17:45 – 18:10 Featured Demo 15
Werner-Körte
Portable Agentic openEO: From Natural Language to Executable Earth Observation Workflows - EURAC Research Italy
Presented by Suriyah Dhinakaran

17:45 – 18:10 Featured Demo 16
Rudolf-Virchow
AKD Labs: An End-to-End Agent Co-Design Studio - The University of Alabama in Huntsville, NASA Marshall Space Flight Center
Presented by Rohit Sisir Sahoo

18:10 – 18:35 Featured Demo 17
Werner-Körte
terraAI: Conversational Earth Observation Analysis - Vilnius University
Presented by Evgenij Shapovalov, Artiom Hovhannisyan, Valentas Gružauskas

18:10 – 18:35 Featured Demo 18
Rudolf-Virchow
Generative Geospatial User Interfaces: Dynamic Dashboard and Storytelling Generation via Agent-to-UI Streaming - EOX IT Services GmbH
Presented by Lubomír Doležal, Josef Prenner

18:35 – 19:00 Featured Demo 19
Werner-Körte
Operating an EO Constellation with AI Agents - Aistralys
Presented by Alexandru Moloceniuc

18:35 – 19:00 Featured Demo 20
Rudolf-Virchow
FLAP-0: An Interactive Platform for Agentic Earth Observation - Barcelona Supercomputing Center and Universitat Politècnica de Catalunya
Presented by Leonidas Kosmidis, Iker Fernandez Andonegui, Marc Solé i Bonet

Day 3 • 21 October 2026

08:00–09:00 *Welcome Coffee*
Foyer & Wandelhalle

09:00–09:45 Keynote: Building Robust Economic Capabilities in AI Agents
Lecture Hall • *Davide Crapis*
Ethereum Foundation
Chair: Dr. Andrzej Kucik, ESA

09:45–10:30 Keynote: Why Mature Agents Require a Paradigm Shift in Tooling
Lecture Hall • *Dr. Christopher Forrest Brown*
Google DeepMind
Chair: Dr. Andrzej Kucik, ESA

10:30–11:00 *Coffee Break | 2 Demos*
Foyer: Demo 18 & Demo 19
Foyer & Wandelhalle

Demo 18

Generative Geospatial User Interfaces: Dynamic Dashboard and Storytelling Generation via Agent-to-UI Streaming
Lubomír Doležal, Josef Prenner • EOX IT Services GmbH

Demo 19

Operating an EO Constellation with AI Agents
Alexandru Moloceniuc • Aistralys

10:30–11:00 *Coffee Break | 2 Demos*
Demo 10 & Demo 16
Expertisium

Demo 10

BlazerOps: Reinforcement Learning Environment for Accelerated Spatial ML Development
Ashin Alex, Harshil Magan • hyades.space

Demo 16

AKD Labs: An End-to-End Agent Co-Design Studio
Rohit Sisir Sahoo • The University of Alabama in Huntsville, NASA Marshall Space Flight Center

10:30–11:00 Featured Demo 21
Werner-Körte
A Live Tour of EVE: An Agentic EO Platform Built on an Open MCP Tool Registry – Pi School
Presented by Àlex R. Atrio, Antonio Lopez, Jino Rohit, Emanuele Micieli

10:30–11:00 Featured Demo 22
Rudolf-Virchow
Agentic AI-Enabled Climate Downscaling with NASA's Prithvi WxC Foundation Model – JPL/NASA, University of Utah
Presented by Hugo Lee, Aashish Panta

11:00–12:00 Oral Session 4: Multi-Modal Large Models, Datasets and Agentic Earth Intelligence
Lecture Hall
Chair: Prof. Dr. Yifang Ban, KTH
This session focuses on the scientific foundations that enable agentic EO systems: Vision-language models, geospatial foundation models, embeddings, semantic reasoning, multimodal grounding, and automated scientific discovery.
This session addresses: "How can foundation models become

reasoning engines for Earth Intelligence?"

11:00–11:12

APERTURE: Semantic Change Understanding and Grounded GeoAI Reasoning via SAR-Optical-Language Alignment

Magdalena Kaiser, Paul Weinmann • *LiveEO*

11:12–11:24

BigEarthNet.txt: A Large-Scale Multi-Sensor Image-Text Dataset and Benchmark for Earth Observation

Johann-Ludwig Herzog, Mathis Jürgen Adler • *BIFOLD and Technische Universität Berlin*

11:24–11:36

Toward Agentic and Unified Agricultural Field Boundary Delineation from Satellite Imagery using Geospatial Foundation Models, Pre-Trained Segmentation, and Embeddings

Sayantan Majumdar • *Division of Hydrologic Sciences, Desert Research Institute*

11:36–11:48

Evaluation of Agentic Autoresearch in an Earth Observation Benchmark

Alex López-Cifuentes • *Xoople*

11:48–12:00

GEOFMOPS: An Agentic Framework for the Geospatial Foundation Model Lifecycle in Earth Observation

Fabio Calefato (University of Bari), Nicolo' Taggio (Planetek Italia), Filippo Lanubile (University of Bari), Fabrizio Rosmarino (University of Bari)

12:15–12:30	Closing Ceremony **Lecture Hall**
12:30–14:00	*Hosted Lunch* ***Foyer & Wandelhalle***
14:00–15:30	Track 1: Building and Operating EO Agents - Building Agentic Earth Observation Workflows with EVE Workbench **Expertisium** • *Andreas Haida, Bara Safrankova - Frank Reply*
14:00–15:30	Track 2: Trustworthy and Scientific Agentic AI - How Do You Know Your Agent Works? Evaluating Earth Observation Agents with Existing Benchmarks **B. von Langenbeck** • *Àlex R. Atrio, Antonio Lopez, Jino Rohit - Pi School, Nicolas Longépé - ESA labs*
14:00–15:30	Track 3: Foundation Models, Embeddings and Lightweight AI (Part 1) - Small Models, Big Maps - Local Geospatial Workflows with Fine-tuned SLMs **Rudolf-Virchow** • *Soumya Ranjan, Daniel Wiesmann - Development Seed*
15:30–16:00	*Coffee Break* ***Foyer & Wandelhalle***
16:00–17:30	Track 1: Building and Operating EO Agents - Agentic Analysis of Earth Observation for Utility Intelligence **Expertisium** • *Fuxun Yu, Rishi Madhok - TerraByte AI*

16:00–17:30	Track 2: Trustworthy and Scientific Agentic AI - From Vibe Coding to Verifiable Scientific Discovery: Agentic Workflows for Earth Observation **B. von Langenbeck** • *Rocco Sedona, Pinaki Banerjee, Alexandre Strube - Forschungszentrum Jülich, Rohit Sahoo - The University of Alabama in Huntsville, NASA Marshall Space Flight Center*
16:00–17:30	Track 3: Foundation Models, Embeddings and Lightweight AI (Part 2) - Small Models, Big Maps - Local Geospatial Workflows with Fine-tuned SLMs **Rudolf-Virchow** • *Soumya Ranjan, Daniel Wiesmann - Development Seed*